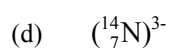
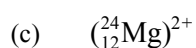
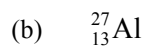


**YEAR 10 SCIENCE EXTENSION
NATURAL AND PROCESSED MATERIALS
ASSIGNMENT 1: ATOMIC STRUCTURE AND EQUATIONS**

NAME: _____

Total: 54

1. Draw the electron shell configuration for each of the following.



(6)

2. Using $({}_{17}^{35}\text{Cl})^{1-}$ as an example, explain the following.(a) Were electrons gained or lost to form the Cl^{1-} ion?

(1)

(b) How many electrons were gained or lost?

(1)

(c) In which group is Cl?

(1)

(d) What valency do all members of this group possess?

(1)

(e) Explain carefully why the $({}_{17}^{35}\text{Cl})^{1-}$ ion is stable, referring to the Octet Rule.

(3)

3. Complete the following table.

NAME	FORMULA
	CO ₂
aluminium sulphate	
dinitrogen tetroxide	
calcium hydroxide	
	(NH ₄) ₂ CO ₃
copper (II) chloride	
	NO ₂
potassium nitrate	
	ZnS
sodium phosphate	

(5)

4. Complete the following table.

ELEMENT/ION	PROTON NUMBER	NEUTRON NUMBER	ELECTRON NUMBER
$^{24}_{12}\text{Mg}$			
$(^{35}_{17}\text{Cl})^{1-}$			
$^{56}_{26}\text{Fe}$			
$(^{27}_{13}\text{Al})^{3+}$			
$^{16}_8\text{O}$			

(5)

5. Using clear and simple diagrams to help, explain how the following bonds arise.

(a) ionic

(b) covalent

(c) metallic

(6)

6. Draw *electron dot diagrams* for the following compounds.

(a) MgCl_2

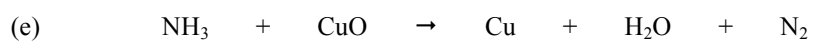
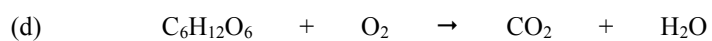
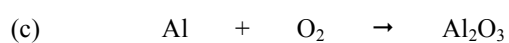
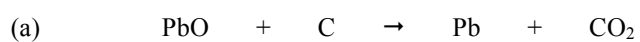
(b) SO_3

(c) NF_3

(d) K_2O

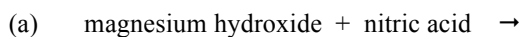
(6)

7. Balance the following equations.



(5)

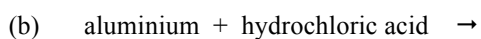
8. Write **complete word equations** for the following acid reactions. Underneath each, write **a balanced molecular equation** and a **balanced ionic equation**.



Molecular:

Ionic:

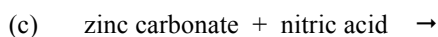
(3)



Molecular:

Ionic:

(3)

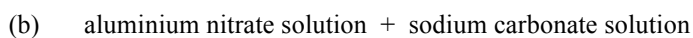
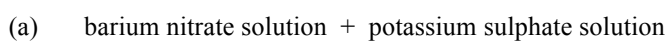


Molecular:

Ionic:

(3)

9. Predict whether a precipitate is produced when the following solutions are mixed together. For those that produce a precipitate, write a **balanced ionic equation**. Write **"no reaction"** if a precipitate is not produced.



(5)